conceivable laws of gravity, say, which ones can exist simultaneously with which conceivable laws of quantum physics that determine the very existence of macroscopic matter? Are all laws we can think of possible, or is there only a restricted number that can somehow be brought into existence? Clearly we have not a glimmering of how to determine which laws of Nature are "possible" and which are not. Nor do we have more than the most rudimentary notion of what correlations of natural laws are "permitted."

For example, Newton's universal law of gravitation specifies that the mutual gravitational force attracting two bodies towards each other is inversely proportional to the square of how far they are apart. You move twice as far from the center of the Earth and you weigh a quarter as much; ten times farther and you weigh only a hundredth of your ordinary weight; etc. It is this inverse square law that permits the exquisite circular and elliptical orbits of planets around the Sun, and moons around the planets—as well as the precision trajectories of our interplanetary spacecraft. If r is the distance between the centers of two masses, we say that the gravitational force varies as $1/r^2$.

But if this exponent were different—if the gravitational law were $1/r^4$, say, rather than $1/r^2$—then the orbits would not close; over billions of revolutions, the planets would spiral in and be consumed in the fiery depths of the Sun, or spiral out and be lost to interstellar space. If the Universe were constructed with an inverse fourth power law rather than an inverse square law, soon there would be no planets for living beings to inhabit.

So of all the possible gravitational force laws, why are we so lucky as to live in a universe sporting a law consistent with life? First, of course, we're so "lucky," because if we weren't, we wouldn't be here to ask the question. It is no mystery that inquisitive beings who evolve on planets can be found only in universes that admit planets. Second, the inverse square law is *not* the only one consistent with stability over billions of years. Any power law less steep than $1/r^3$ ($1/r^{2.99}$ or $1/r$, for example) will keep a planet in the *vicinity* of a circular orbit even if it's given a